# A Comprehensive Edge-Aware UAV–LiDAR Rock Point Cloud Segmentation Approach for Automated Rock Dump Monitoring

Rongjun Xiong, Shaobo Li, Vincent J. L. Gan, *Member, IEEE*, Aiguo Sun, Sheng Du, *Member, IEEE*, and Yunlong Wu

***Abstract*—The accurate rock dump monitoring is a critical task in waterway maintenance. Uncrewed aerial vehicle (UAV)–LiDAR has attracted increasing attention due to its high efficiency and automation potential. However, in large-scale UAV-derived point clouds, challenges, such as significant size disparities among rocks, spatial overlaps, and uneven point density hinder reliable rock area extraction and segmentation, thereby limiting the accurate grain size estimation for monitoring purposes. To address these issues, we first employ designed identification markers to rapidly extract the target rock area. Focusing on the core challenges of rock segmentation, we observe that height variations between adjacent rocks generate distinct depth edges, which serve as reliable cues for structural delineation. Leveraging this characteristic, we propose a novel edge-aware segmentation method. The approach begins with local phase analysis to highlight boundary regions of both large and small rocks, followed by enhancement filtering and edge detection. These extracted edges guide a constrained region-growing process, enabling the coarse segmentation of individual rock units. Subsequently, a global energy optimization step refines the segmentation results by merging fragmented clusters into coherent rock entities, effectively mitigating over-segmentation caused by surface irregularities. The proposed method demonstrates strong robustness to overlapping rocks and complex pile configurations, achieving segmentation accuracy with 85% and 78% F1-scores in two real cases.**

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## I. Introduction

THE large-scale waterway transport faces an enormous challenge to provide an insurance of stable waterway structures against the scouring and washing [1], [2]. Accurate rock dumping is critical for addressing this challenge. Precisely dumping rocks that meet the specifications from the transport ship into the designated underwater area protects the embankment or bank reinforcements. The gradation of the rocks must be well-distributed, and their volume and quality should fully meet the design specifications. Consequently, the monitoring process of rock volume and grain size evaluation becomes extremely important. Traditional rock volume estimation is achieved through employing the "first-batch weighing method" for quantity measurement, while the tapeline measurement is used for grain size determination. This mainly approach relies on the manual visual inspection methods, which are low inefficient and inaccurate. LiDAR provides an automatic way; however, traditional LiDAR scanning based on terrestrial laser scanning (TLS) requires complex multiple station setups when trying to get point clouds covering the whole rocks [3]; robotic-based LiDAR cannot be well operated because of the limited space on the ship [4]. In comparison, uncrewed aerial vehicle (UAV)–LiDAR scanning technology has significant advantages in terms of efficiency and low-manual intervention [5] and can provide reliable scanning from the air while offering a full coverage of the rocks across heterogeneous regions. Thus, it has great potential for future fully automated monitoring applications.

Currently, UAV–LiDAR monitoring of rock-carrying ships has no established precedents and faces several challenges. Despite ongoing algorithmic advances, practical engineering applications are hindered by complex data characteristics and rock structures. The wide scanning range of UAV–LiDAR captures a large number of nonrock points, making rapid detection and separation of ships and rock areas difficult, especially as higher flight altitudes increase scene coverage and complexity [see Fig. 1(a)]. Moreover, in practical applications, it is essential to record rock volume and grain size for each individual ship in every dumping cycle. This raises a key challenge: how to reliably distinguish and identify multiple ships to ensure the accurate data association. Even

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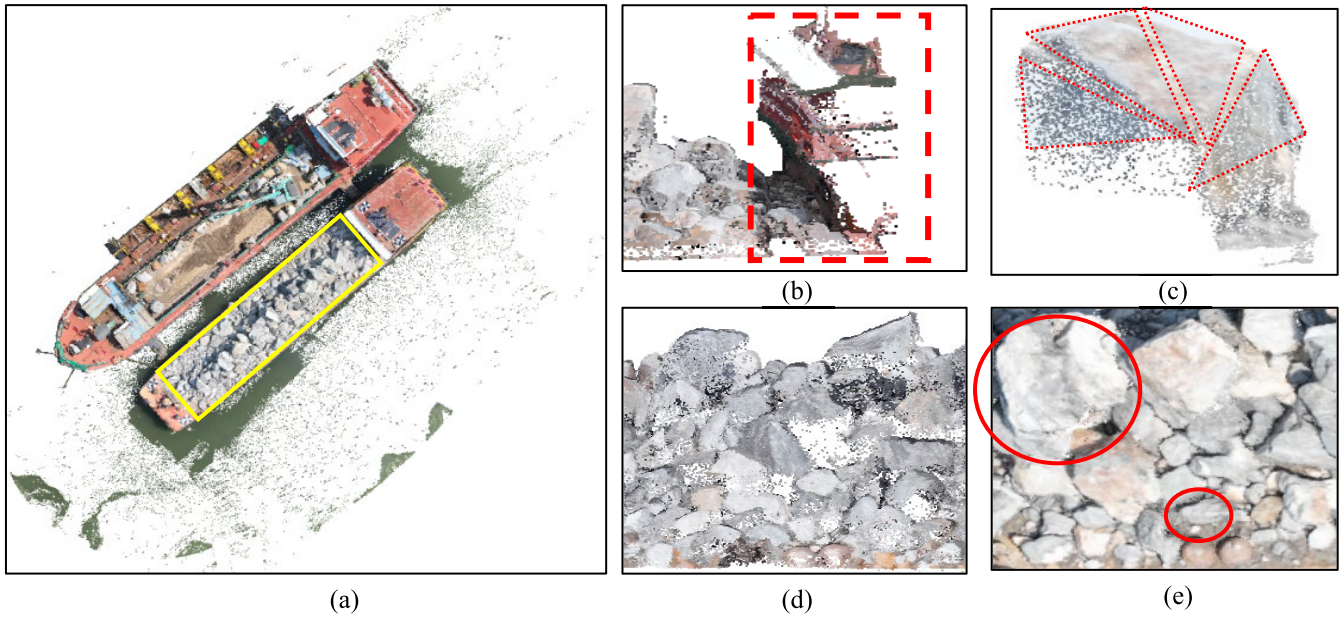



Fig. 1. Data of the UAV–LiDAR-based approach for rock dumping automatic monitoring. (a) Bird view of the surveyed data (yellow rectangle indicates the rock area). (b) Poop deck (red dotted rectangle) and rock of the ship. (c) Surfaces of a rock (rectangles show different surfaces). (d) Overlapping of rocks. (e) Different sizes of rocks, circles indicate the large and small rocks.

when measurement devices are well focused on the rock area for imaging, complex rock structure challenges arise. These challenges include the intrinsic characteristics of the rocks and the unevenness of the point cloud data:

1) Uneven data distribution causes point density to be highly variable on top surfaces and low on sides. This makes choosing an appropriate neighborhood scale difficult: small scales lead to under-segmentation, and large scales lead to over-segmentation. As shown in Fig. 1(c), different surfaces of a rock have very different densities, and the side surface has extremely sparse points. Irregular rock surfaces also hinder reliable feature extraction, as traditional descriptors like normal vector and curvature become unreliable. Shape fitting methods, such as plane fitting [6] or RANSAC [7] struggle with such uneven geometry. Moreover, as shown in Fig. 1(c), a single rock is composed of four separate surfaces, and all of these surfaces have different normal vectors. Traditional strategies may segment them apart, leading to over-segmentation.
2) The natural variability of rock piles leads to differences in grain size and overlapping layouts. Overlapping rocks blur boundaries, especially for small rocks, further complicating segmentation. As shown in Fig. 1(d), rocks of different colors overlap, making it difficult to distinguish each rock. Large and small rocks are hard to segment simultaneously due to their scale differences. The size difference between large and small rocks can be as large as several times, as shown in Fig. 1(e). On small rocks, geometric features like normal, roughness, and curvature [8] are unstable or unrepresentative. Their limited point count makes them prone to being misclassified as noise.

To address these problems, a UAV–LiDAR ship rock monitoring approach is proposed. The main contributions are threefold.

1) A comprehensive framework for identification marker design, detection, and recognition is proposed. The identification marker design and detection facilitate the accurate rock region extraction and support reliable rock segmentation.
2) A combined local phase and enhancement filtering algorithm is proposed to enhance rock edges. Since raw grayscale images poorly reveal small rock continuity and edges, monogenic signal analysis is used to get local phase, which captures the fine structural continuity and is insensitive to intensity variations. However, the local phase is also sensitive to noise. To achieve the accurate and continuous edge detection, an edge structure descriptor based on structure tensor, combining the Frangi filtering and Harris corner detection ideas, is further proposed and employed to enhance edge-like structures.
3) An optimized strategy for rock segmentation is proposed. To address the issue of complex surface distribution of rocks that can lead to over-segmentation, a global energy optimization framework is introduced. A data term model based on intersection over union (IoU) and a neighborhood term model based on triangulated irregular network (TIN) and centroid distance are constructed. Multiple factors are integrated to jointly optimize the optimization process.

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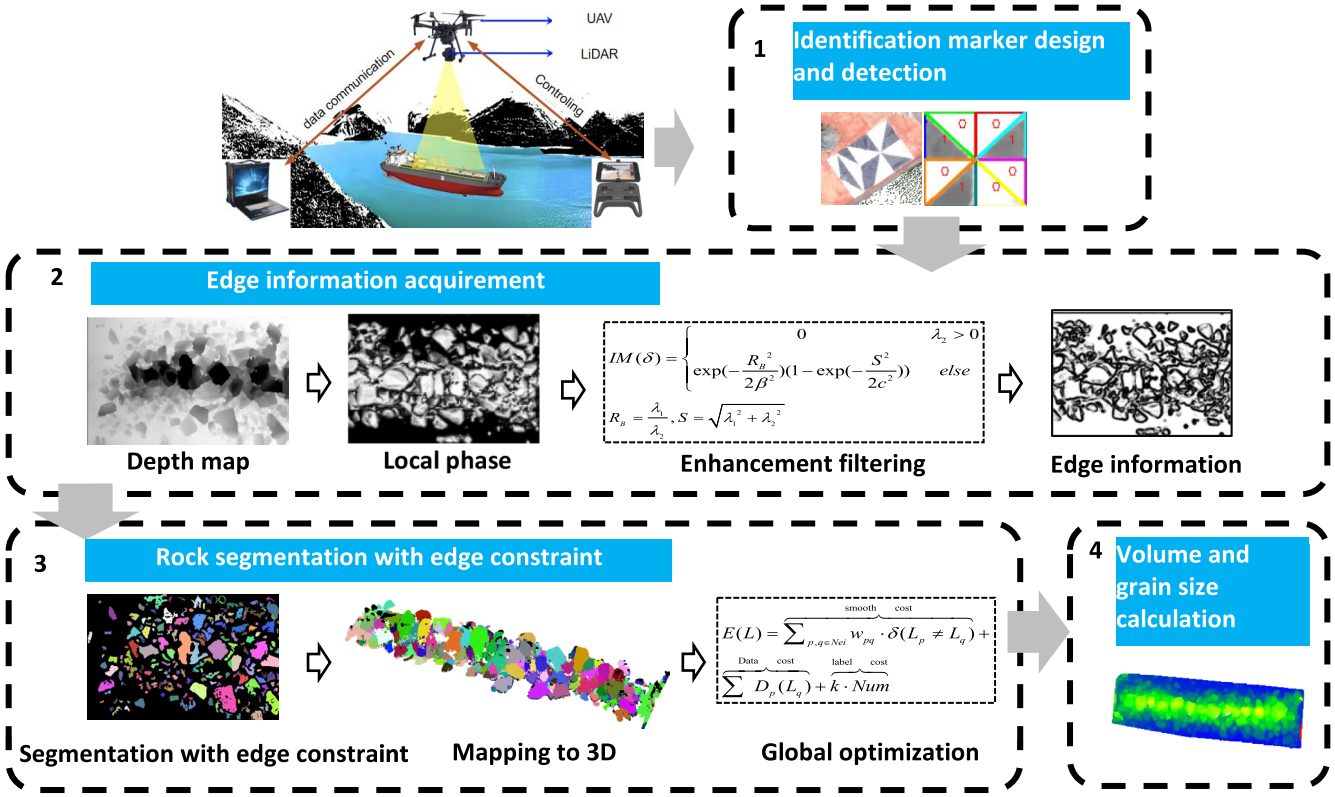



Fig. 2. Flowchart of the proposed method.

This approach represents a practical and effective implementation of photogrammetry and remote sensing techniques, highlighting the strong applicability of UAV–LiDAR technology in real-world engineering contexts. The rest of this article is organized as follows. The related works are introduced in Section II. Section III gives detailed descriptions of the algorithms. Sections IV and V describe and analyze the experimental results and discussions. Finally, our conclusions are drawn in Section VI. The flowchart of the proposed method is shown in Fig. 2.

## II. Related Works

The rock segmentation remains a challenging problem. A comprehensive review of 3-D point cloud segmentation methods is provided by Yang et al. [9]; yet, rock-specific segmentation has not been explicitly addressed. Regarding nondeep learning rock segmentation, Galanakis et al. [10] employed singular value decomposition (SVD) for rock segmentation in structure-from-motion point clouds, but such methods are effective only when joints and rocks can be clearly separated. Similarly, detailed approaches for rock wall instance segmentation in TLS point clouds have been proposed by Valero et al. [11] and implemented in a publicly available toolbox [12]. While these methods can yield accurate results when geometric cues allow for a clear distinction between rocks and joints [13], such conditions are absent in the case of rock dumping, where overlapping geometries obscure boundaries. The geometric characteristics of rocks in our dataset are highly complex, as they are not composed of simple planar or smoothly curved surfaces. This complexity poses significant challenges for extracting discriminative geometric features. Moreover, individual rocks frequently overlap without well-defined boundaries, making their separation difficult. Without reliable rock extraction, accurate grain-size estimation becomes infeasible.

Although deep-learning-based point cloud segmentation methods have recently attracted considerable attention, there are only a few algorithms for rock surface extraction or the division of pebbles based on deep learning [14], [15], [16]. The limited availability of annotated rock datasets hinders the development of a robust training corpus for such frameworks [17], [18]. Edge-based algorithms provide fast segmentation of a point cloud because of their simplicity. The general goal of edge-based methods is to classify pixels with significant local changes in intensity or color. For 3-D point cloud data, these edge points can then be used to determine the shape of objects. Wiedemann and Holst [19] proposed a segmentation algorithm based on geometric features derived from intensity values. There are also other approaches using the intensity attribute for segmentation in structures with rubble masonry [20]. Recently, Daghigh et al. [21] proposed a watershed-based segmentation algorithm using 3-D point clouds and reflection intensity images, which can be adopted to identify individual rocks in rubble masonry dams and applied for deformation monitoring. However, the intensity value varies with the object, detection range, and weather and radiometric distortion [21]. It is difficult to correct the intensity value to obtain relatively true intensity data, especially for UAV-based

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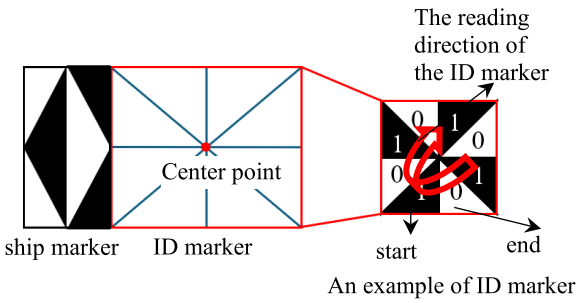



Fig. 3. Identification marker design.

LiDAR scans. Walicka and Pfeifer [22] proposed a method for instance segmentation of individual grains from a TLS point cloud representing a mountain riverbed. Their method is only suitable for large rocks with clear spaces between individual rocks, rather overlapping rocks.

In summary, existing methods struggle to handle the dense overlapping of rocks, as well as the simultaneous segmentation of large and small rocks within the same scene. In this study, we specifically use the depth map derived from point clouds for segmentation. The differences between the rocks appear as the changes in height, and the edges in depth are precisely caused by these height variations. Therefore, using the edge information of the depth map for auxiliary segmentation is very beneficial.

## III. Algorithms

### A. Preparation and Preprocessing Before Segmentation

*1) Identification Marker Design and Detection:* The accurate detection of rock-carrying ships is challenged by limited labeled data and the need to link rock properties to individual ships. To address this, we propose a marker-based detection and recognition framework. The system uses high-contrast black-and-white markers comprising two parts: a ship marker for detection and an ID marker for unique identification (as shown in Fig. 3). The ship marker features a black arrowhead pointing toward the bow and interlaced triangular patterns to enhance geometric texture. This design ensures reliable detection in point clouds and distinguishes ships from natural and artificial surroundings. To enable individual ship identification and link rock properties to specific ships, unique IDs are encoded in black-and-white triangular markers. Each marker is a square panel divided into eight isosceles right triangles by diagonals and mid-lines. Each triangle is set to black (1) or white (0), generating $28 = 256$ unique binary patterns. Starting from the lower left and proceeding clockwise, the pattern forms an 8-bit ID; e.g., 10101010 = 170. Four markers are placed at the corners of each ship's cargo area in a rectangular layout for multiview visibility and spatial registration. Based on the LiDAR resolution, markers require a minimum size of $0.8 \times 1.2$ m for reliable detection; thus, a standard size of $1.0 \times 1.5$ m is adopted.

After deploying the identification markers and completing the UAV–LiDAR measurement, we provide a process for marker detection and recognition. First, the cloth simulation filtering (CSF) [23] is applied to the flipped data to effectively remove the water surface points first. Then, we build a bird-view image of UAV–LiDAR data for identification marker detection and recognition. Projecting the measured point cloud data onto the X–Y plane after setting the grid resolution yields the corresponding image $f$. The resulting image is similar to that in Fig. 1(a). Ship marker detection is performed using the template matching. A reference template $t$ (the ship marker in Fig. 3) is slid and rotated across $f$ to find the best match. The normalized cross-correlation (NCC) coefficient is computed at each position $(u, v)$, defined as follows [24]:

$$\gamma = \frac{\sum_{x,y} \left(f(x,y) - \bar{f}_{u,v}\right)\left(t(x-u, y-v) - \bar{t}\right)}{\sqrt{\sum_{x,y} \left(f(x,y) - \bar{f}_{u,v}\right)^2 \sum_{x,y} \left(t(x-u, y-v) - \bar{t}\right)^2}} \tag{1}$$

where $\bar{t}$ and $\bar{f}_{u,v}$ are the mean intensities of the template and corresponding image region, respectively. NCC is robust to brightness and contrast changes. To accelerate the computation, integral maps of $f(x, y)$ and their squared values are precomputed, enabling the fast calculation of local sums and variances [25], [26]. This significantly improves efficiency, especially for multiscale or multiorientation searches. The marker's location is determined by the position $(u, v)$ with the highest NCC score.

Using the geometric relationship to the ship marker, the ID marker is located behind it. Despite noise and illumination variations, the ID marker, composed of eight isosceles right triangles, is divided accordingly, and the mean gray level of each triangle is computed. K-means clustering groups these eight values into two classes: higher intensities as "white" (0) and lower as "black" (1). The binary sequence is then read clockwise from the lower left triangle (see Fig. 3) to decode the ship ID. For example, starting from the lower left and proceeding clockwise, the pattern forms an 8-bit ID; e.g., 10101010 = 170.

*2) Processing for the Extraction of the Rock Area:* To enable the dumping monitoring, the spatial extent of the rock distribution must be extracted first, as subsequent volume computation and segmentation are performed exclusively on the rock-bearing point cloud. Based on the known positions of the four identification markers put on the ship, a coarse rock area is initially delineated by excluding points lying outside the quadrilateral formed by these markers. However, this preliminary region may still include points from nonrock structures, such as the ship's poop deck [as shown in Fig. 1(b)]. To reduce interference, we propose a depth map-based refinement method. The coarse point cloud is projected into a 2-D depth map, where higher elevations appear brighter. K-means clustering divides the map into low-, medium-, and high-intensity regions. Columns where high-intensity pixels exceed 70% are identified as the ship's poop deck and removed. Finally, distance-based clustering retains only the largest connected component, the main rock pile, for subsequent volume and grain size analysis.

### B. Rock Point Cloud Segmentation

After the above process delineates the rock distribution area, the next step is to segment the rock point cloud to support the

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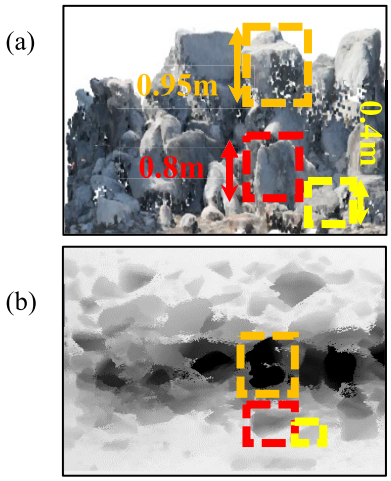



Fig. 4. Illustration of edges of rock characteristics. (a) Rocks with different height differences (orange, red, and yellow dotted rectangles indicate the rocks with height differences: 0.95, 0.8, and 0.4 m). (b) Corresponding area in depth map [orange, red, and yellow dotted rectangles indicate the same area in (a)].

dumping monitoring. Traditional rock segmentation methods typically rely on the extraction of local geometric features (e.g., normal vectors and curvature [8], [27]), followed by a region-growing process that merges neighboring points based on geometric similarity [28]. However, these approaches face challenges due to the subtle interrock differences, the irregular and complex morphology of rock surfaces, and the presence of uneven point density caused by partial occlusion or nonuniform LiDAR scanning.

To overcome these limitations, we observe that rocks of varying sizes exhibit distinct edges in the depth map (as shown in Fig. 4), even in challenging cases such as small rocks (marked by the yellow dotted rectangle) and overlapping rock groups (indicated by orange and red dotted rectangles). These edges remain clearly discernible due to significant height discontinuities at rock boundaries. Leveraging this characteristic, we propose an edge-aware segmentation strategy that integrates local phase calculation, enhancement filtering, edge extraction, and a global energy optimization process to generate coherent rock point cloud clusters. The strong contrast in elevation at rock perimeters enhances edge saliency, providing critical cues for accurate rock separation.

*1) Local Phase Calculation:* The size differences between large and small rocks can be as large as several times. Consequently, edges in the depth map have different gray levels since the depth differences between rocks vary. The usual adopted threshold-based algorithms cannot be directly applied to the depth map for extraction; otherwise, serious over-extraction and under-extraction will occur. One possible way is to highlight all the edges to make the threshold selection simple. The local phase is introduced in this article to achieve this goal [as shown in Fig. 5(b)], which is derived from monogenic signal analysis and can be employed to highlight local structures since it removes the influences of intensity variations.

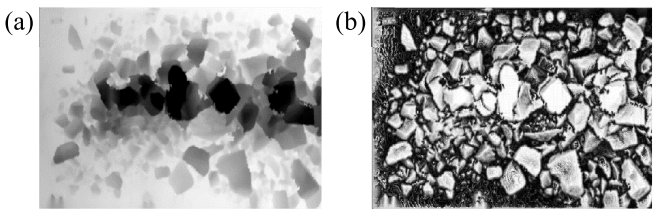



Fig. 5. Illustration of the local phase image. (a) Rocks in depth map image. (b) Corresponding local phase image which highlights all the rock edges.

For a 2-D signal $f(\mathbf{x})$ ($\mathbf{x} = [x_1, x_2]^{\mathrm{T}}$), the combination of a signal $f(\mathbf{x})$ with its transform forms a 2-D generalization of the analytic signal, and the monogenic signal is obtained through [29], [30]

$$f_M(x_1, x_2) = (b^* f)(x_1, x_2) + (h_1{}^* b^* f)(x_1, x_2) + (h_2{}^* b^* f)(x_1, x_2) \tag{2}$$

where $f_M(\mathbf{x})$ is the monogenic signal corresponding to $f(\mathbf{x})$. $b$ is the time domain representation of $B$, which is the bandpass filter with the transfer function and is used as follows:

$$B(\chi) = \chi \cdot \exp\left(-\chi^2 \cdot wl^2\right) \tag{3}$$

where $wl$ is the center wavelength expressed in pixels. $h_1$ and $h_2$ represent the spatial representation of the convolutional kernel of the Riesz transform

$$(h_1(x_1, x_2), h_2(x_1, x_2)) = \left(-\frac{x_1}{2\pi\left(x_1^2 + x_2^2\right)^{\frac{3}{2}}}, -\frac{x_2}{2\pi\left(x_1^2 + x_2^2\right)^{\frac{3}{2}}}\right). \tag{4}$$

Setting $f_{sig} = b^* f$, the local phase is then obtained using

$$\Phi(x_1, x_2) = \operatorname{atan}\left(\frac{|R\{f_{sig}\}(x_1, x_2)|}{f_{sig}(x_1, x_2)}\right). \tag{5}$$

For our work, we set the depth map as a 2-D signal, namely, $f(\mathbf{x})$. By applying (2)–(5) and converting $\Phi$ to a range of 0–255, the local phase image $LP$ is obtained.

*2) Edge Enhancement Filtering Algorithm:* The phase image $LP$ has highlighted all the obvious structures. To better extract the edge structures in $LP$ from a large number of nonedge pixels, a depth map-based edge enhancement algorithm is presented.

First, the 2-D structure tensor is introduced that is derived from the local gray-level variation analysis [31]. The 2-D structure tensor $\mathbf{M}$ is a symmetric positive matrix, and its corresponding eigen-decomposition yields two real eigenvalues and two orthogonal eigenvectors [as shown in (6)], which are helpful for local structure description

$$\mathbf{M} = G_\delta * \begin{bmatrix} I_x^2 & I_x I_y \\ I_x I_y & I_y^2 \end{bmatrix} \tag{6}$$

where $I_x$ and $I_y$ are the first derivatives of the $LP$ along $x$- and $y$-directions (row-column directions), respectively. $G_\delta$ is a Gaussian filter with the scale parameter set as $\delta$. The eigenvalues can be used to indicate the local structures, which are achieved through

$$\det(\mathbf{M} - \lambda \mathbf{E}) = 0 \tag{7}$$

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where det(·) denotes the determinant and **E** is the identity matrix. $\lambda_2$ ($\lambda_1 \geq \lambda_2$) are the eigenvalues of matrix **H**. If and $\lambda_2$ are small, it means that the analyzed region is a flat area. If one eigenvalue is high and the other low, it means that there are small gray level changes in one direction and a significant change in the orthogonal direction [32], which indicates that there is an edge-like structure. If $\lambda_1$ and are high, it means that the gray level changes in any direction and indicates that there is a corner point.

For image structure description, two well-known algorithms are the Frangi filtering [32] and the Harris corner detector [31]. Frangi's method enhances line-like structures using eigenvalues of the Hessian matrix but is not especially designed for edge detection. The Harris method, based on the trace and determinant of the structure tensor, is mainly used for corner detection. Based on the above analysis, we design the following filter to enhance local edge structures by combining the Frangi filtering and Harris corner detection ideas:

$$IM(\delta) = \begin{cases} 0, & \lambda_2 > 0 \\ \exp\left(-\dfrac{R_B^2}{2\beta^2}\right)\left(1-\exp\left(-\dfrac{S^2}{2c^2}\right)\right), & else \end{cases}$$
$$R_B = \frac{\lambda_1}{\lambda_2}, S = \sqrt{\lambda_1^2+\lambda_2^2} \tag{8}$$

where IM($\delta$) is the filtered image when setting the scale parameter as the Gaussian filter scale $\delta$. $R_B$ is an indicator for edge structure, while there is an edge structure, $R_B$ will be high. $S$ is adopted for indicating high gray-level areas. The sizes of $S$ and $R_B$ are the same as the input image. $\beta$ and $c$ are the parameters. The ratio of $S$ attains its maximum for a significant structure and is 0, whenever $\lambda_1$ and $\lambda_2$ tend to vanish. The ratio $R_B$ accounts for the aspect ratio of the eigenvalues. This ratio is essential for distinguishing between edge structures.

The process of the above enhancement filtering is summarized as follows.

1) Calculate the first derivatives of the depth image and Gaussian filter in (6). This is essentially equivalent to the first-order derivative function of the Gaussian filter convolved with the depth map.
2) Based on the matrix **H**, compute the eigenvalues and eigenvectors. Then, construct the edge structure descriptor $R_B$ and $S$ in (8).
3) Set different $\delta$ to obtain different enhancement-filtered *IM* data using (8). Extract the maximum *IM* under different scale parameters as the final result.

In summary, we compute the local phase of the depth map to obtain a local phase image. Then, the edge enhancement algorithm mentioned above is applied to the local phase image to generate an enhanced image, in which edge information is further highlighted. Finally, the edge extraction is performed based on the enhanced image. Then, the Otsu algorithm [33] is used for edge extraction. An edge image with the edge setting as 255, and the other area setting as 0 can be obtained. An edge map $\boldsymbol{E}_{\mathrm{m}}$ is obtained.

*3) Region Growing With the Edge Constraint:* By extracting these edges based on the edge-enhanced depth map, closed boundary structures can be established around individual rocks, which constrain intracluster connections and preventing the over-merging of neighboring clusters, even in the presence of complex rock surfaces. Here, the region-growing algorithm is adopted to divide the point cloud dataset $\boldsymbol{A} = \{p_1(x_1, y_1, i_1), p_2(x_2, y_2, i_2),..., p_{\mathrm{m}}(x_m, y_m, i_m)\}$ (where $i_i$ denotes the color values of $p_i$ point) into $n$ point cloud clusters $\boldsymbol{C}\{p_1,p_2,..., p_m\}$ of different attributes [6], [34], and the clustering process needs to consider the edge constraint, which means that when the region grows to an edge, then the growing across the edge direction will be ended. The steps are shown as follows.

1) *Neighborhood Search:* For each point, find all neighbors within radius $r$.
2) *Seed Initialization:* Start with an empty seed set; select an unsegmented, nonedge point as the initial seed.
3) *Region Growing:* For seed point $P0$, check unsegmented neighbors within $r$. If the height difference $d \leq Td$ (a height threshold) and the neighbor is not an edge point in *Em*, add it to the seed set. Mark $P0$ as segmented after evaluation.

Repeat until all points are labeled, forming a 2-D cluster map with unique colors per cluster. Map pixels back to 3-D points for full segmentation. Remaining edge points are assigned to the nearest cluster if their height is below the cluster's average and horizontal distance is minimal. The final output, $\Psi$, contains distinct rock clusters. This edge-aware method uses sharp elevation drops in the depth map to guide the segmentation, enabling the robust separation of weakly connected and overlapping rocks.

*4) Refinement Using Global Energy Optimization:* The above steps basically achieve the segmentation of rock point clouds. However, a single rock is composed of several separate surfaces which may lead to different edges within a single rock causing over-segmentation. Thus, further processing is needed. Setting the primary segmentation result as basic units $\Psi$, and each basic unit is a set of points segmented as the same rock. The global energy optimization formula is employed here for basic unit refinement, and it is set as follows [6]:

$$E(L) = \overbrace{\sum_{p,q\in Nei} w_{pq}\cdot\delta\left(L_p \neq L_q\right)}^{\text{smooth cost}} + \overbrace{\sum_{D_p(L_q)}}^{\text{Data cost}} + \overbrace{k\cdot Num}^{\text{label cost}} \tag{9}$$

where $L$ is a given set of labels for basic units and $\delta$ is an indicator function. The optimization task is to assign a basic unit $p\epsilon\Psi$ a label $L_q\epsilon L$ such that the labeling $L$ minimizes the energy $E(L)$, where *Num* is the minimum number of points required for a valid rock and *Nei* is a predefined neighborhood for basic units. $w_{pq}$ measures the centroid distance between two basic units.

The data cost term $D_p(L_q)$ measures the sum of geometric error between the basic unit $p$ and $q$. We took into account that if small rocks were part of a larger rock, their spatial areas would necessarily overlap. Thus, we directly extract the cube boundaries of basic units and adopt the IoU of the two

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Fig. 6. Conduction of the UAV–LiDAR-based approach for rock dumping automatic monitoring. (a) Workers putting the identification marker on the ship. (b) Measurement of the rock manually. (c) UAV–LiDAR scanning above the ship.

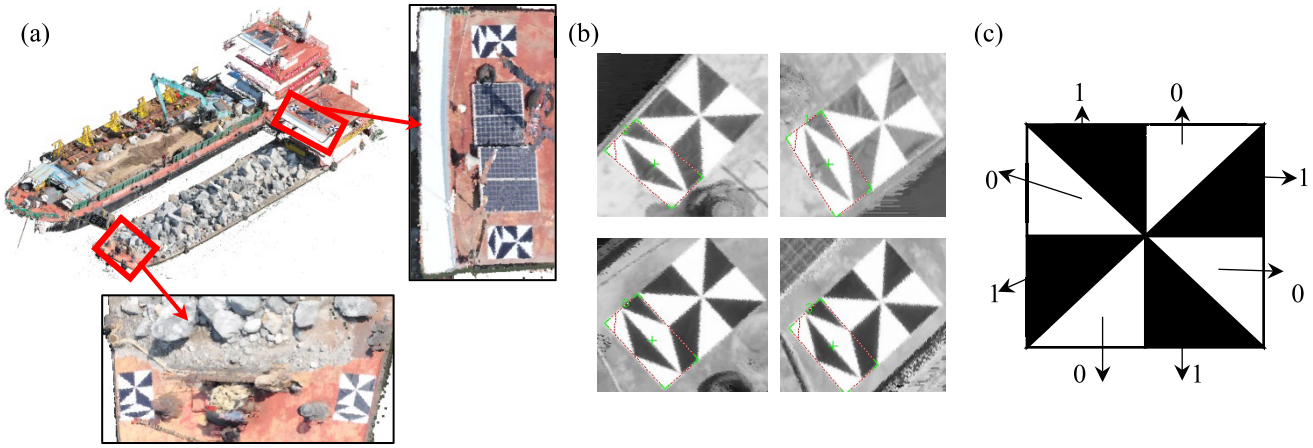



Fig. 7. Data with markers. (a) Rock data after CSF processing. Red rectangle areas show the marker on the ship. (b) Detection of the four identification ship markers (red dotted lines indicate the detected ship marker). (c) ID number of the marker.

boundaries as the data cost term. This means that a larger IoU promotes the two basic units merged. The IoU can be obtained as follows [35]:

$$IoU = \frac{|P \cap Q|}{|P \cup Q|}. \tag{10}$$

where $P$ and $Q$ are the cube bounding boxes of two basic units.

The neighbor system *Nei* is constructed by extracting the centroid point of each basic unit as nodes and building a TIN using these nodes. Then, the neighboring relationship between two basic units is described by the node adjacency relationship and distance. The minimization of (9) can be computed through the extended $\alpha$-expansion method. After the above optimization process, the final segmentation result can be obtained.

### C. Volume Calculation and Grain Size Estimation Based on Segmentation Result

After segmentation, principal component analysis (PCA) is applied to each point cloud cluster in the segmented result. The resulting eigenvalues are used to approximate the lengths of the major axes of the individual rocks, from which the average axis length per cluster is computed. This average serves as a proxy for grain size, enabling a rough estimation of rock dimensions.

For rock volume computation, both the rock surface and the underlying deck surface must be determined to calculate their vertical elevation difference. Given the deck's near-planar nature, its surface is estimated using the extracted deck edge points. We first identify planar segments by fitting $a$ plane to each cluster in the segmented result and retaining those with fitting residuals below $3\sigma$ (where $\sigma$ is the standard deviation of the fit). These planar clusters are aggregated and fit with a quadratic polynomial to model the global deck surface, accounting for minor curvature or tilt. A regular grid is then established over the rock distribution area. The deck elevation at each grid node is computed from the fit polynomial, while the rock surface is interpolated using inverse distance weighting [36]. Finally, the volume is calculated as the sum of the products of the elevation differences between the two surfaces and the area of each grid cell.

## IV. Experiments

Excavator-based dumping is typically adopted in shallow water with relatively low-current velocity, especially in nearshore and gentle slope areas. The data were collected by the DJI L2 series and then processed using DJI Terrain to generate the point cloud data. The adopted LiDAR has a maximum of 240 000 pts/s in single-return mode and 1200 000 pts/s in multiple-return mode. The horizontal resolution is 5 cm, and the vertical resolution is 4 cm when the

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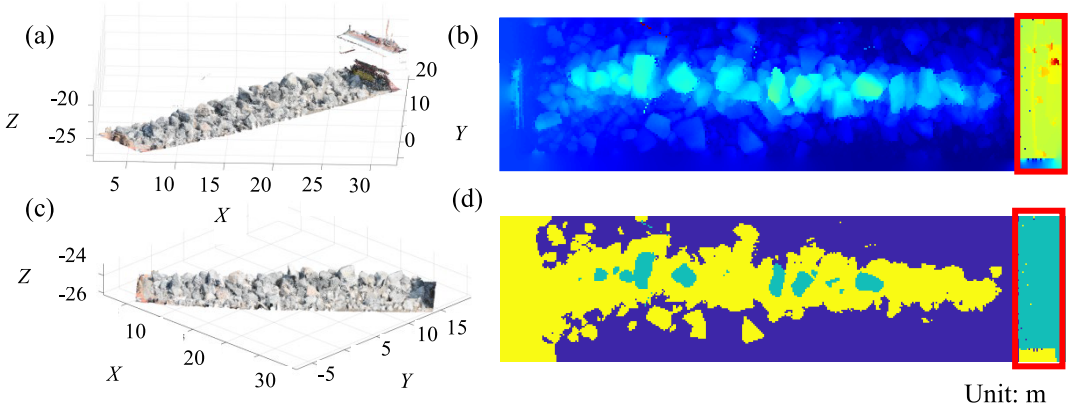



Fig. 8. Rock area extraction result. (a) Point cloud of the ship. (b) Corresponding depth map of (a). (c) Final extracted rock area. (d) k-means clustering result based on (b), and the red rectangle shows the area that should be removed.

detection range is 150 m. The repetitive scanning pattern has a horizontal angle of 70° and a vertical angle of 3°. During the calculation process, we chose the highest point cloud imaging resolution, point cloud accuracy refinement, as well as filtering to ensure that the identification markers were clearly imaged. During surveying, we first put the identification markers on the ship [as shown in Fig. 6(a)], and also measured the length of several rocks, as shown in Fig. 6(b). The UAV–LiDAR scan is illustrated in Fig. 6(c).

The identification markers in Fig. 7 are clear, with the front part facilitating easy ship detection, as shown by the red rectangle and detection scores above 0.87. The ship ID is read from the back part, which consists of eight triangles divided into "white" (0) and "black" (1) based on their intensity. By traversing these, the ID 170 is recognized, allowing all rock survey data from this ship to be categorized under this ID. Using the known positions of the identification markers as the four corners of a bounding rectangle, the ship's main body is extracted by removing points outside this region, as shown in Fig. 8(a). However, the elevated poop deck, included due to marker placement, has higher elevation and risks misclassification as rock, which would skew volume estimates. To remove it, a depth map is generated by projecting the point cloud onto a 2-D grid, with z-values encoded as grayscale, as shown in Fig. 8(b). The bright region (red rectangle) indicates the high-elevation poop deck. K-means clustering divides the depth map into three elevation levels (low, medium, and high); the high cluster marks the poop deck [bluish green in Fig. 8(d)]. For each vertical column, if over 70% of pixels belong to the high cluster, the entire column is flagged for removal. This mask is applied to the 3-D point cloud, effectively eliminating poop deck points. The cleaned result [see Fig. 8(c)] retains only the main deck for accurate downstream processing.

A rock segmentation algorithm was then adopted. First, we transformed the rock point cloud into a depth map, as shown in Fig. 9(a). To highlight the edges among different rocks, local phase, and edge enhancement filtering were applied, as illustrated in Fig. 9(b) and (c). The red rectangles in Fig. 9(a) indicate regions with weak edge features. However, after local phase calculation, these regions exhibited enhanced intensity and were well separated from the background. The ellipse in Fig. 9(a) shows an overlapping case, where a small rock lies on top of a larger one. In this scenario, the edge between the two rocks is not sufficiently distinct, which could hinder segmentation if applied directly. After local phase calculation and enhancement filtering, however, the two rocks were effectively separated, as shown in Fig. 9(b) and (c). In summary, closed boundary structures were well established around individual rocks, constraining intracluster connections and preventing the over-merging of neighboring clusters, even on complex rock surfaces. Subsequently, the edge points were used as the growth stop criterion. After region growing, the result in Fig. 9(e) was obtained. However, the points in the edge areas were not segmented. To address this, we considered the spatial relationship between the edge points and the segmented points to classify the edge points into different clusters. The final segmentation result is in Fig. 9(f). We selected several individual segmented rocks, as shown in Fig. 10. The main structures were correctly segmented, and the spatial features were preserved, which further validates the performance of our method.

Each individual rock possesses a distinct grain size. In practical operations, grain size is typically measured manually. In this study, we compare rock sizes derived from manual measurements with those computed using the proposed automated method. PCA is applied to each segmented rock cluster, and the length of the major axis obtained from PCA is used as a proxy for rock size. These size estimates are aggregated to construct a grain size distribution histogram, as shown in Fig. 11(a). The grain size results are also consistent with those of several rocks that we measured manually. To evaluate the segmentation accuracy of the proposed method, a manual segmentation of the point cloud was performed, treating each individual rock as a separate entity. The result is presented in Fig. 11(b), where different colors represent distinct rocks. During manual annotation, small outliers and residual deck points were carefully removed to ensure a clean reference dataset. Quantitative evaluation is conducted using standard metrics: precision, recall, and F1-score [37]. To align the automated segmentation results with the manual reference, the centroids of rock clusters in both datasets are computed. A pair of rocks, one from the manual result and one from

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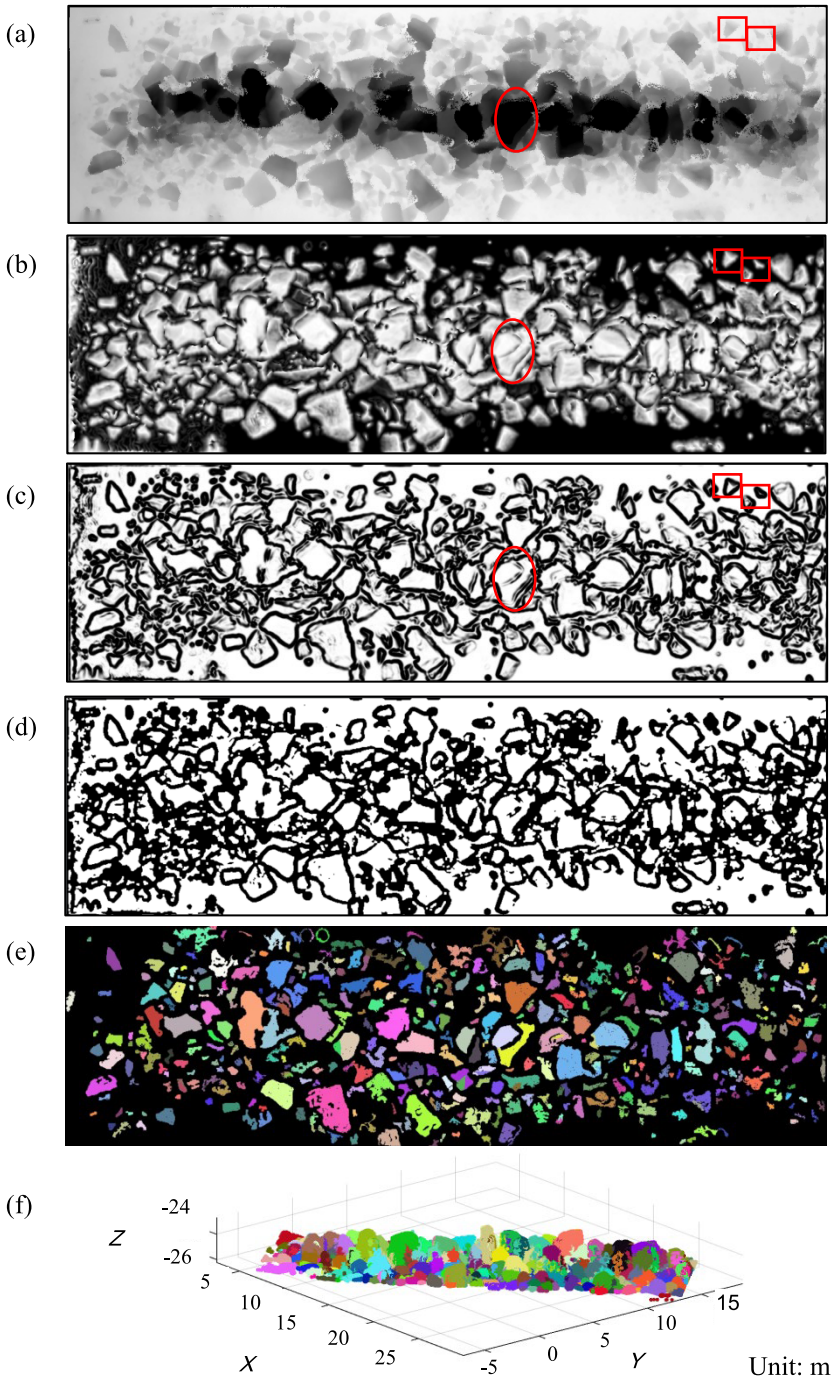



Fig. 9. Rock segmentation result using the proposed method. (a) New generated depth map of rocks. (b) Obtained local phase data. (c) Enhanced edge of (b). (d) Edge data. (e) Segmentation result with the constraint of (d). (f) Final segmentation result.

the automated result, is considered a match if the Euclidean distance between their centroid is less than 0.5 m. There are 390 rocks in the automatic segmented result aligned with the manual result. The number of manual results is 463. There are 73-point cloud rocks missed in our result, and 66 rocks in the automatic result have no corresponding manual ones. 82 rocks were over-segmented in the automatic result. The final performance of the proposed method is: precision of 0.86, recall of 0.84, and F1-score of 0.85, as shown in Table I.

The rock volume in the paper is calculated through building the grid model and computing the surface model. Planar-like structural points were also extracted, as shown in Fig. 12(a), with the majority corresponding to the deck edge regions. These points were used to fit a continuous surface, which serves as an approximation for the underlying deck surface, as depicted in Fig. 12(b). The final rock volume was computed by integrating the elevation differences between the rock surface grid [see Fig. 12(c)] and the interpolated deck surface grid. In this case, the computed volume is 353 $m^3$, which is in close

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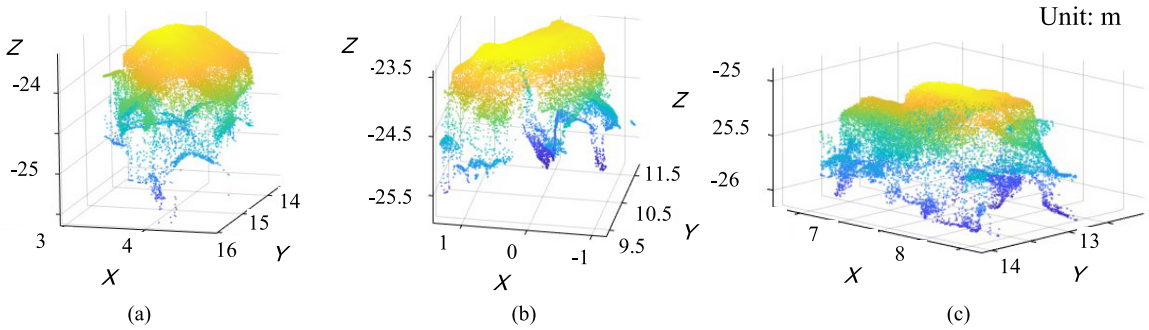



Fig. 10. Several rock point clouds. (a)–(c) Different rocks.

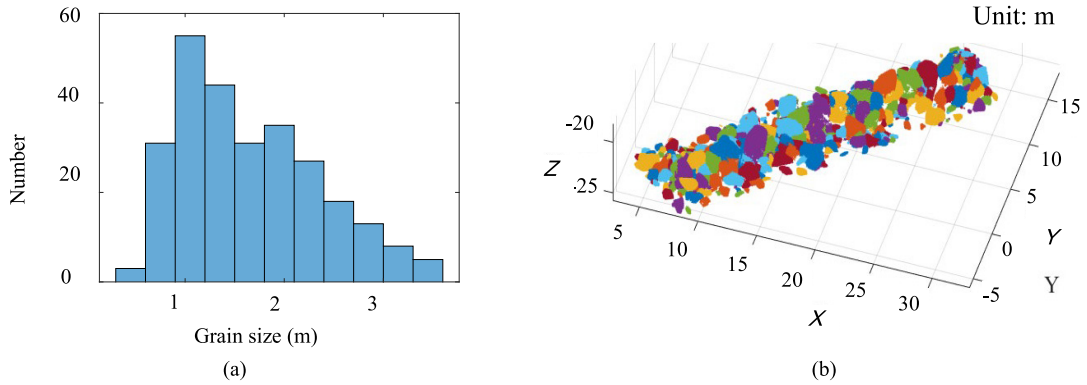



Fig. 11. Grain size distribution and ground truth of rock segmentation. (a) Histogram of segmented rocks' grain sizes. (b) Ground truth of rocks.

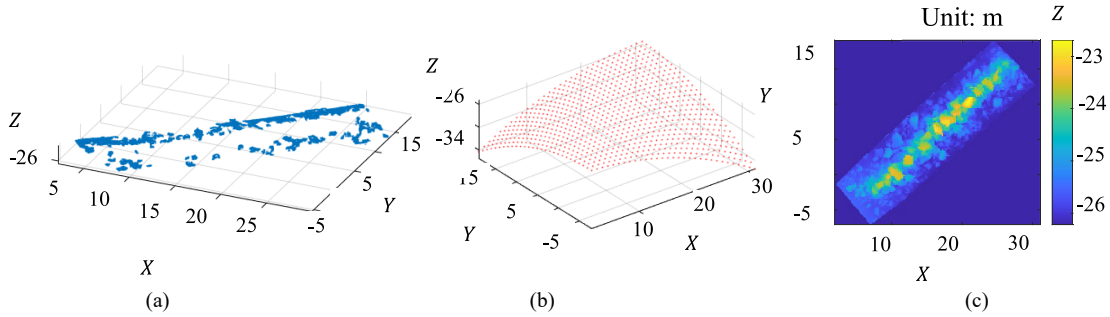



Fig. 12. Data for calculating the rock volume. (a) Extracted deck data. (b) Fit deck surface. (c) Grid model of the rock.

TABLE I
QUANTITATIVE ANALYSIS RESULTS OF DATA1

| F1-score | Precision | Recall | Relative bias of volume |
|---|---|---|---|
| 85% | 86% | 84% | 2.8% |

agreement with the close-range LiDAR station measurement of 363 $m^3$ (used as the ground truth), demonstrating the reliability of the proposed approach with a relative difference of approximately 2.8%.

## V. DISCUSSION

### A. Ablation Experiment

*1) Without Edge Constraint:* Edge constraint is a crucial component of the proposed segmentation work. To evaluate its effectiveness, we removed this constraint and directly applied the region growth based on depth maps, comparing the performance with and without the edge constraint while keeping all other processes unchanged. The labeled segmented map without edge constraint is shown in Fig. 13(a), and the final result is shown in Fig. 13(b). Caused by the irregular rock structures, there is serious over-segmentation resulting in poor segmentation performance [white rectangle area in Fig. 13(a)]. This phenomenon is mainly induced by the fact that the segmentation is interrupted by the variations of geometric structures, and it is hard to keep the integrity of one rock.

After statistics, there are 447 rocks in automatic segmented result aligned with the manual results, whose number is 463. There are 16 point cloud rocks missed in the result, and 541 rocks in the automatic result have no corresponding manual ones. The 45 rocks were over-segmented in the automatic result. The final performance of this result is: precision of 0.45, recall of 0.97, and F1-score of 0.62, which is much lower than that of the proposed method. The result is shown in Table II.

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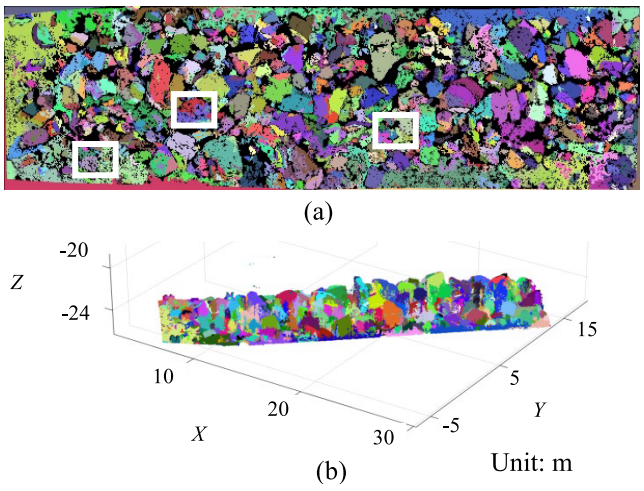



Fig. 13. Segmentation without edge constraint. (a) Segmented result of depth map (white rectangles indicate over-segmentation). (b) Final segmented point clouds.

TABLE II
QUANTITATIVE ANALYSIS RESULTS OF ABLATION EXPERIMENTS

| Method | F1-score | Precision | Recall |
|---|---|---|---|
| Ablation experiment 1 | 62% | 45% | 97% |
| Ablation experiment 2 | 82% | 81% | 83% |
| Ablation experiment 3 | 71% | 77% | 67% |
| Ablation experiment 4 | 83% | 78% | 90% |

*2) Using Proposed Enhancement Filtering Only:* To evaluate the effectiveness of the proposed local phase and global energy optimization processes, we retained only the enhancement filtering while removing the other processes, keeping all remaining steps the same. In this step, the obtained edge map is shown in Fig. 14(a). There are 386 rocks in the automatically segmented result aligned with the manual result. The number of manual results is 463. There are 78 point cloud rocks missed in our result, and 91 rocks in the automatic result have no corresponding manual ones. The 60 rocks were over-segmented in the automatic result. The final performance is: precision of 0.81, recall of 0.83, and F1-score of 0.82. The result is relatively lower than the proposed approach. It is also noted that most of the rocks have been segmented well, which indicates the superiority of using the edge information as a constraint. The result is shown in Table II.

*3) Using Frangi Enhancement Filtering Only:* The proposed enhancement filtering algorithm was also compared with the Frangi enhancement filter, a widely used method for enhancing tubular structures that can also be applied to depth map edges. In this comparison, we replaced our edge enhancement processing with a Frangi filter-based algorithm, thus omitting both the phase information and the global energy optimization. As shown in Fig. 15(a), the Frangi filter-based edge extraction successfully extracts many edge pixels; however, the edges are often interrupted, and those corresponding to individual rocks frequently fail to form closed

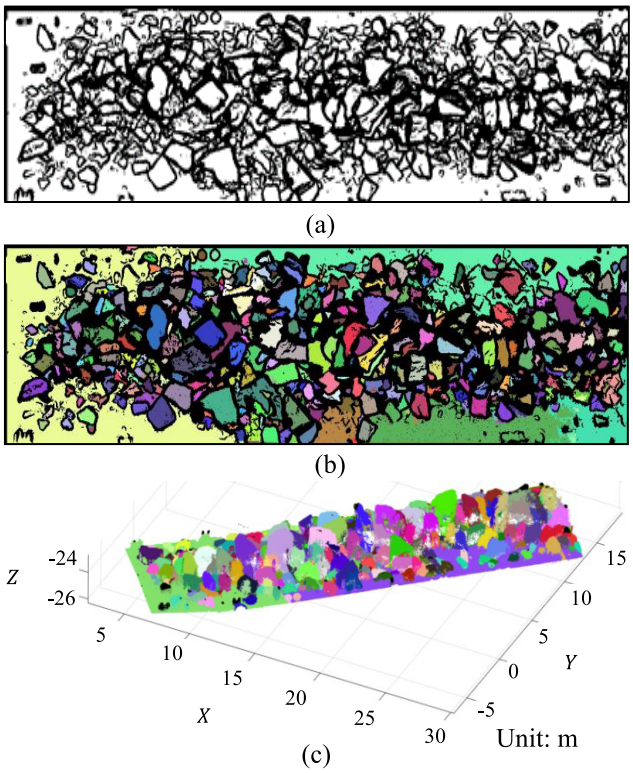



Fig. 14. Rock segmentation result using the proposed enhancement filtering and without phase information as well as global energy optimization. (a) Generated edge map of rocks. (b) Segmentation result with the constraint of (a). (c) Final segmentation result.

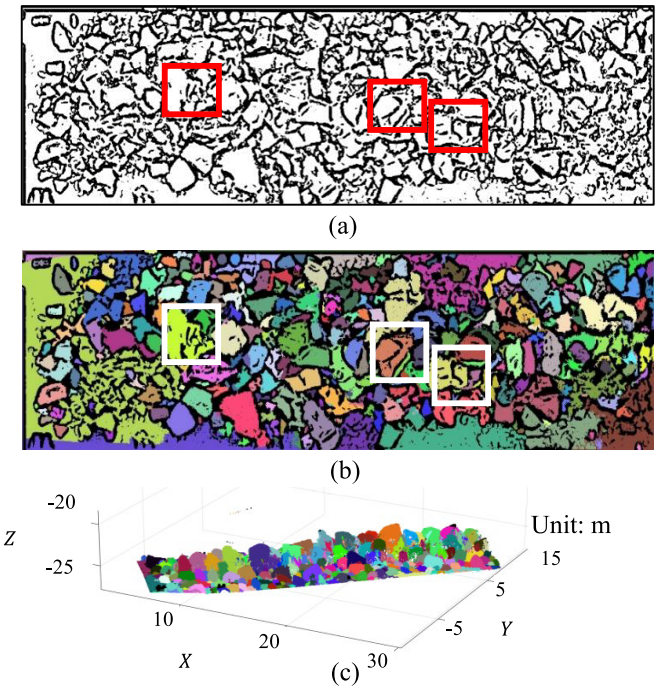



Fig. 15. Segmentation with the Frangi edge constraint. (a) Frangi edge of depth map (red rectangles indicate under-segmentation). (b) Segmentation image based on depth map with Frangi edge constraint (white rectangles indicate under-segmentation). (c) Final 3-D segmentation result.

boundaries [indicated by red rectangles in Fig. 15(a) and white rectangles in Fig. 15(b)]. This limitation arises from the Frangi filter's reliance on line-like structural characteristics,

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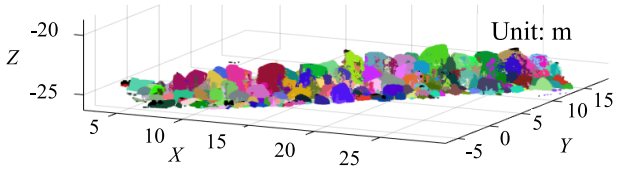



Fig. 16. Segmentation with phase information and without global energy optimization.

whereas rock edges do not always conform to such shapes. Consequently, during region growing, the interrupted edges in Fig. 15(a) cannot provide effective constraints, leading to under-segmentation [white rectangle areas in Fig. 15(b)] in complex areas, where simple, nonedge-constrained methods also perform poorly. The final segmented point clouds are presented in Fig. 15(c).

There are 308 rocks in the automatic segmented result aligned with the manual result. There are 155-point cloud rocks missed in our result, and 93 rocks in the automatic result have no corresponding manual ones. 59 rocks were over-segmented in the automatic result. The final performance of this result is: precision of 0.77, recall of 0.67, and F1-score of 0.71 (shown in Table II), which is lower than that of the proposed method.

*4) Effect of Phase Information (Excluding Global Energy Optimization):* The global energy optimization process is a very useful part of reducing the over-segmentation. To evaluate the effectiveness of this section, we removed this part to compare the performance. The final result is shown in Fig. 16. There are 415 rocks in automatic segmented result aligned with the manual results, whose number is 463. There are 48 point cloud rocks missed in the result, and 117 rocks in the automatic result have no corresponding manual ones. The 77 rocks were over-segmented in the automatic result. The final performance of this result is: precision of 0.78, recall of 0.90, and F1-score of 0.83, which is lower than that of the proposed method. Affected by the irregular rock surface structures, there is serious over-segmentation resulting in poor segmentation performance. This phenomenon is mainly caused by the fact that the segmentation is interrupted by variations in geometric structures, and that it is hard to maintain the integrity of a single rock, leading to poor precision and F1-score, as shown in Table II.

### B. Result Analysis of Another Dataset

We also used another dataset to verify the presented method. The data extracted based on identification markers is shown in Fig. 17(a), and we refined the result considering the depth map information. The extracted rock area is shown in Fig. 17(b).

The generated depth map is shown in Fig. 18(a), and the extracted edges are shown in Fig. 18(c). In the right portion of the data, some rocks have very small exposed surfaces, whereas their corresponding edges occupy relatively large regions. As a result, the segmented image contains sizable unsegmented areas. Nevertheless, by rearranging these unsegmented points, the final segmentation result remains highly consistent with the ground truth, as illustrated in Fig. 18(d).

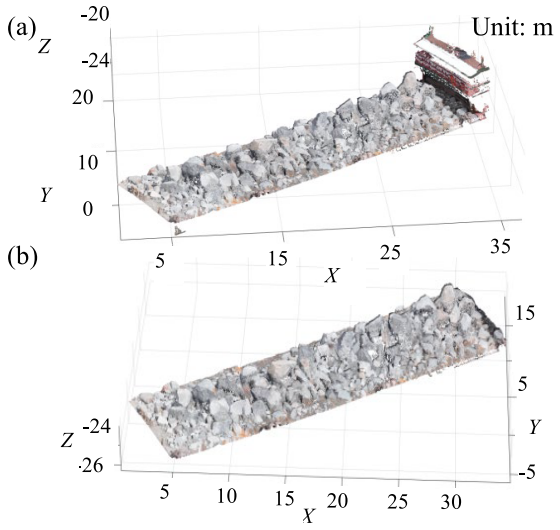



Fig. 17. Rock area extraction result. (a) Rock area extracted with identification markers constraint. (b) Final extracted rock result.

TABLE III
QUANTITATIVE ANALYSIS RESULTS OF DATA 2

| F1-score | Precision | Recall | Relative bias of volume |
|---|---|---|---|
| 78% | 73% | 84% | 3.2% |

There are 493 rocks in the automatic segmented result aligned with the manual results, whose number is 382. There are 58 point cloud rocks missed in our result, and 122 rocks in the automatic result have no corresponding manual ones. The 51 rocks were over-segmented in the automatic result. The final performance of the proposed method is: precision of 0.73, recall of 0.84, and F1-score of 0.78, as shown in Table III. The performance of 0.78 still indicates good applicability and generalization capability of the proposed method. We used the major axis from PCA results to represent the size parameters of rocks. All the size parameters can be used to construct a histogram and are shown in Fig. 19(a). Plane-like structural points were extracted, as shown in Fig. 19(b), with the dominant component corresponding to the peripheral regions of the ship deck. These points were fit to a continuous surface, representing the underlying deck beneath the rock pile [see Fig. 19(c)]. The rock volume was then obtained by integrating the elevation differences between the rock surface grid and the fit deck surface grid. The volume is 450 $m^3$, matching the ground truth of 436 $m^3$, with a relative error of 3.2%.

### C. Compared With Deep-Learning-Based Method

Currently, there is no suitable point cloud segmentation algorithm for single rock instance segmentation based on the point cloud data. To achieve the deep-learning-based automatic segmentation of rock points, the 3-D point cloud data is first projected onto a 2-D plane image, and YOLOv11 was then applied. The projection process is based on the X–Y plane projection of the point cloud, normalizing the point cloud coordinates to pixel coordinates at a set spatial resolution, while

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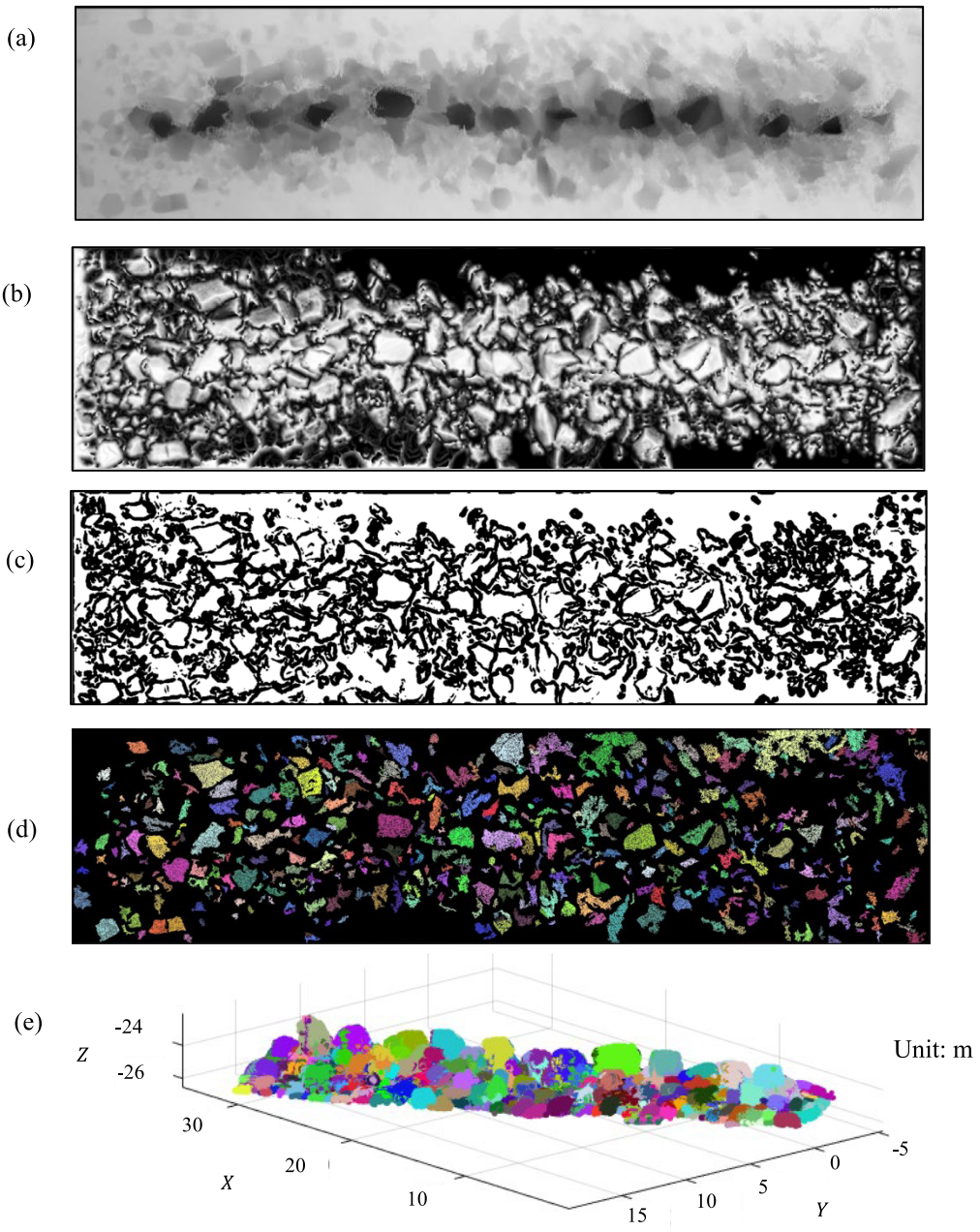



Fig. 18. Rock segmentation result using the proposed method. (a) New generated depth map of rocks. (b) Obtained local phase data. (c) Edge data. (d) Segmentation result with the constraint of (c). (e) Final segmentation result.

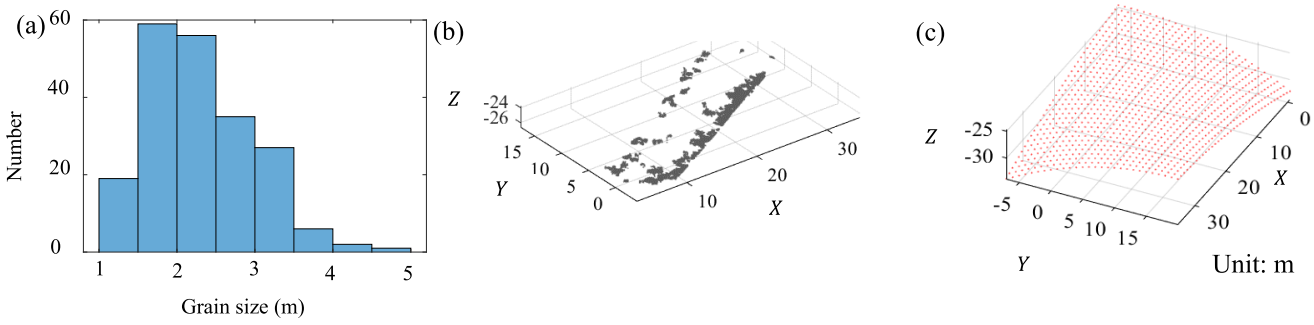



Fig. 19. Data for calculating the rock volume. (a) Histogram of rock sizes. (b) Extracted deck data. (c) Fit deck surface.

retaining the RGB information to generate the corresponding color image. This image is used as the input for model training and prediction. During the annotation stage, the open-source annotation tool is used to perform polygon instance annotation

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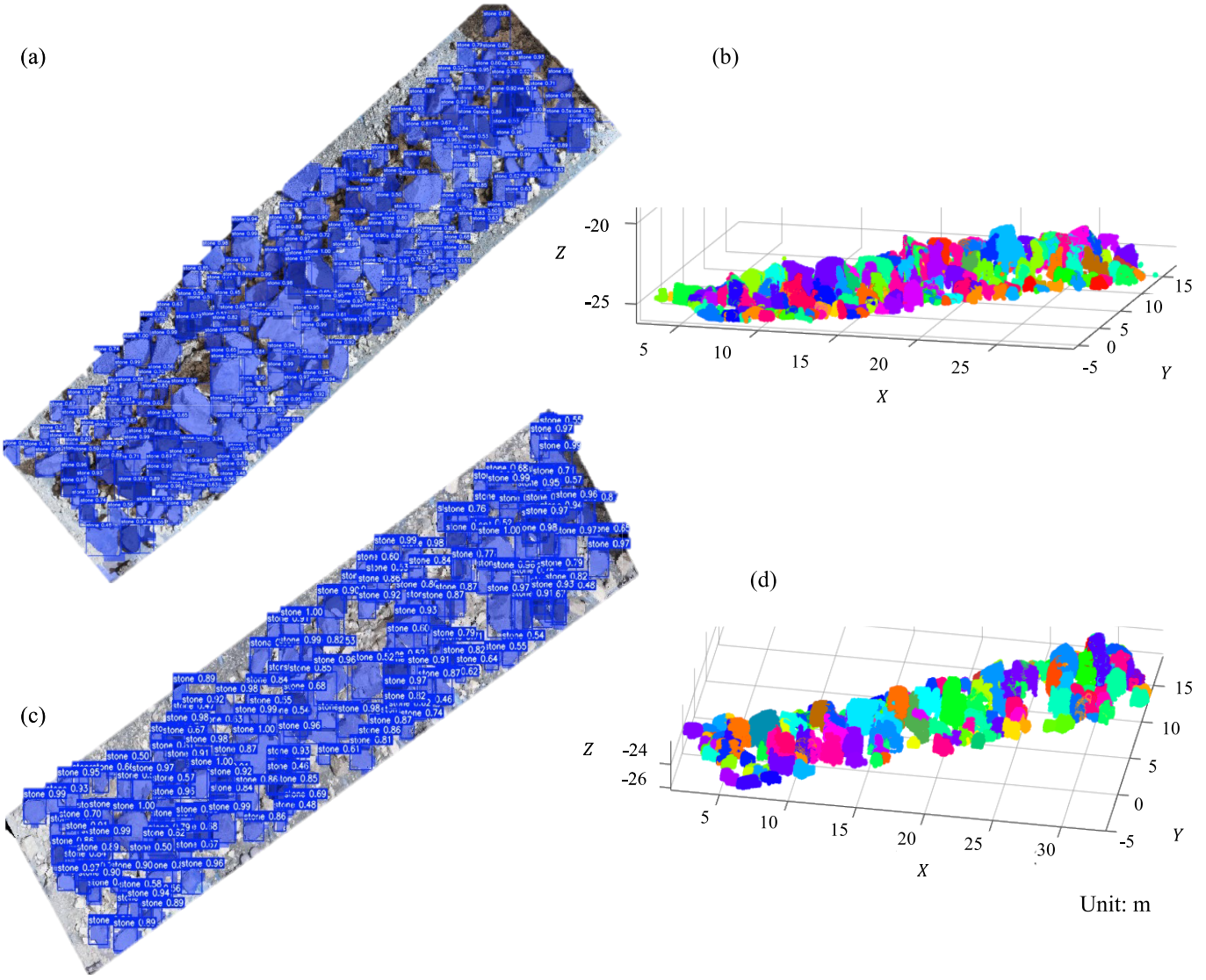



Fig. 20. Recognition result using YOLOv11. (a) Result of data 1. (b) 3-D version of (a). (c) Result of data 2. (d) 3-D version of (c). Different rectangles in (a) and (c) mark different rocks, and in each rectangle, the rock has been segmented.

on the target regions in the projected image. A total of 1008 rock instances were annotated, with 695 for training and 313 for validation. The annotation results were saved in YOLO format, with each instance recorded in the form of normalized polygon mask coordinates, facilitating the direct integration with the model training process. The final result is shown in Fig. 20.

In the training stage, the YOLOv11 instance model is used as the base network. The model input is the color image obtained through projection, and the label file contains the corresponding polygon mask coordinates. After learning on the training and validation sets, the model outputs the weight file. In the prediction stage, the instance segmentation results are output in the YOLO standard format. The segmentation results are then remapped to the 3-D point cloud through the aforementioned inverse projection algorithm, achieving the automatic separation and extraction of each rock target in the point cloud space.

TABLE IV
QUANTITATIVE ANALYSIS RESULTS OF THE PROPOSED METHOD AND THE YOLO METHOD

| | *Precision* | *Recall* | *F1 Score* | *Time cost* |
|---|---|---|---|---|
| Our method in data 1 | 86% | 84% | **85%** | 18.7 s |
| Deep learning in data 1 | 82% | 60% | 69% | <1 s |
| Our method in data 2 | 73% | 84% | **78%** | 15.2 s |
| Deep learning in data 2 | 88% | 49% | 63% | <1 s |

We applied the trained model to the two datasets directly. In the first dataset, there are 279 rocks in the automatic segmented result aligned with the manual result. The number of manual results is 463. There are 184 point cloud rocks missed in our result, and 61 rocks in the automatic result

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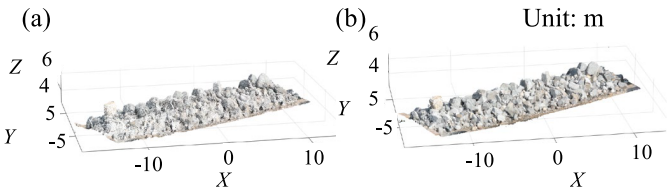



Fig. 21. Data with different densities. (a) High-density data. (b) Low-density data.

have no corresponding manual ones. The 56 rocks were over-segmented in the automatic result. The final performance of the deep learning-based method is: precision of 0.82, recall of 0.60, and F1-score of 0.70. In the second dataset, there are 187 rocks in automatic segmented result aligned with the manual result. The number of manual results is 382. There are 195 point cloud rocks missed in our result, and 25 rocks in the automatic result have no corresponding manual ones. 32 rocks were over-segmented in the automatic result. The final performance of the deep learning-based method is: precision of 0.88, recall of 0.49, and F1-score of 0.63. The comparison result is shown in Table IV.

Both of the performance is lower than the proposed method. In summary, the YOLOv11-based algorithm has picked large rocks well, but it also missed lots of small rocks, leading to low recall values. We also compared the time cost of the proposed method with that of the deep learning-based method. Excluding the time required for marker detection and bird-view image generation, we recorded only the rock segmentation time. The results are presented in Table III. The deep-learning-based method is notably fast.

Although the proposed method requires more time, it remains sufficiently efficient to meet the demands of field operations.

### D. Effect of the Point Density

Point density does have an impact on the segmentation result in three aspects: 1) decreasing the saliency of the rock edges; 2) introducing data voids in the depth map; and 3) ignoring small rocks. To explore the impact of the point density in real cases, we conducted a flight experiment with different UAV flight paths to obtain different datasets, rather than downsampling the existing data. After the flight, two datasets were obtained. The density of the first dataset is 12 955 points/m$^2$, while the second one is 901 points/m$^2$. The rock area data from the two datasets are shown in Fig. 21. After applying our method to the two datasets, the corresponding depth map segmentation results are shown in Fig. 22. After the energy optimization, the final 3-D segmentation results are shown in Fig. 23.

Rock edges need dense points to describe the edge features, and therefore the method performs better on dense data. As shown in the red rectangles in Fig. 23, a serious over-segmentation occurs in Fig. 23(b) marked by the red rectangle. However, it is surprising that the result based on the sparse data still achieved a relatively satisfactory performance, albeit barely so. It is noted that the sparse data size is only 7% of the dense dataset, while the F1-score is only 6% lower than that

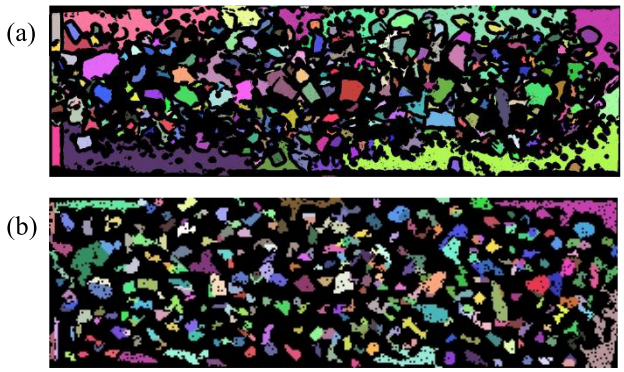



Fig. 22. Depth map segmentation results of high-density data and low-density data. (a) Segmentation result of high-density data. (b) Segmentation result of low-density data.

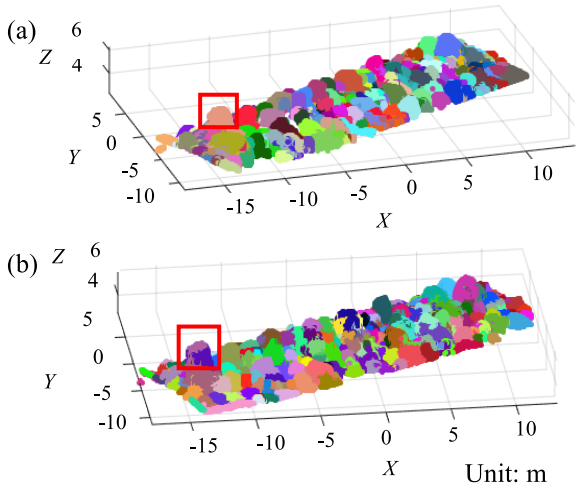



Fig. 23. Point cloud segmentation results of high-density data and low-density data. (a) Segmentation result of high-density data. (b) Segmentation result of low-density data.

TABLE V
QUANTITATIVE ANALYSIS RESULTS OF HIGH-DENSITY DATA AND LOW-DENSITY DATA

| | F1-score | Precision | Recall |
|---|---|---|---|
| High density data | 76% | 84% | 70% |
| Low density data | 70% | 81% | 61% |

of the result based on the dense data, as shown in Table V. This analysis indicates that the proposed method even has the potential for real-time data transmission and calculation, since real-time operations always have the data density limitation.

### E. Effect of the Data Quality

The quality of LiDAR points is vital. In fact, for UAV–LiDAR surveying, a complex simultaneous localization and mapping (SLAM) is integrated into the system to obtain the accurate position information by combining data from images, point clouds, and Global Navigation Satellite System (GNSS). While surveying an area without distinctive landmarks, the performance of SLAM may be degraded, leading to poor point cloud quality. Here, we adopted another dataset with poor imaging performance. The data are shown in

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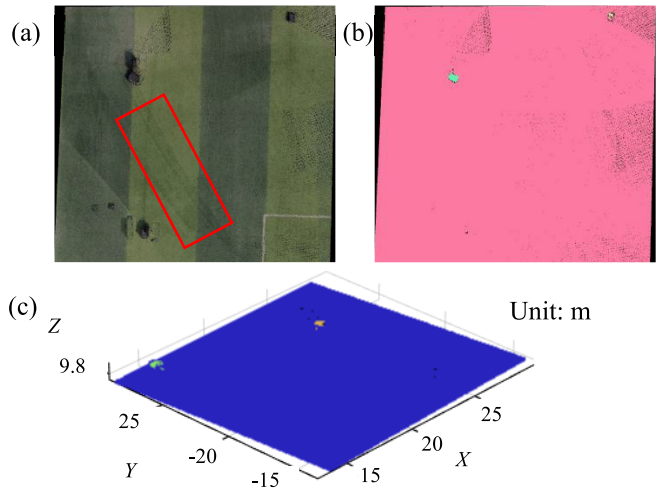



Fig. 24. Result based on a relatively poor UAV–LiDAR survey. (a) Bird's-eye view of the data. (b) Depth map segmentation result based on the data. (c) Final 3-D segmentation based on the data. The red rectangle indicates the ghosting phenomenon.

Fig. 24. It is observed that there is a ghosting phenomenon in Fig. 24(a) marked by the red rectangle, which may be caused by poor position information. Three objects are located on the playground, while the edges of the objects are blurred. The final segmentation results are shown in Fig. 24(c). Only two objects were segmented. This experiment indicates that point data quality does have an important impact on the presented algorithm. However, in most cases, better hardware choices and more distinctive environments are capable of generating a satisfactory point cloud, leading to a better segmentation result.

## VI. Conclusion

This article has presented a robust edge-aware segmentation framework for automated rock dump monitoring using UAV–LiDAR data. Our study demonstrated that depth map edge information facilitates the separation of distinct rocks. To leverage this, we introduced a local phase calculation and developed an edge-aware enhancement filtering algorithm, which effectively highlights rock boundaries across various scales and occlusion levels. Consequently, the precise segmentation was realized and subsequently refined through global energy optimization. A key advantage of the proposed method is its computational efficiency, stemming from its reliance on depth map computation rather than intensive point cloud processing.

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